

Problem Statement #61 – Household Carbon Footprint & Target Tracker

SRN	PES1UG25CS846
Student Name	SIDDU SANGANNA SHIRASAGI
Target Actors	Resident, Sustainability Coach

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest household monthly electricity (kWh) and transport (km) logs to compute total monthly carbon equivalent (CO ₂ e) emissions.	High	Pass: Calculated emissions match the specified EPA conversion-factor formulas for the entered electricity and transport values. Fail: calculated emissions are zero or negative when valid positive consumption is entered.	Core function for measuring household carbon footprint.
FR-002	Functional	The system shall allow a Resident to record monthly household gas consumption and commute kilometres along with the electricity consumption for a selected month.	High	Pass: A valid month can be saved with electricity, gas, and commute values, and the saved values are displayed when the same month is reopened. Fail: the system saves an incomplete or invalid record.	Provides the monthly activity data required for footprint tracking.
FR-003	Functional	The system shall display the Resident's monthly CO ₂ e emissions and historical trend for the recorded months.	Medium	Pass: After a successful calculation, the selected month's CO ₂ e value and the available historical monthly values are displayed in chronological order. Fail: a saved month is missing or shown with a different calculated value.	Allows the Resident to understand changes in emissions over time.
FR-004	Functional	The system shall allow a Resident to set a target CO ₂ e reduction milestone and display progress toward that target.	High	Pass: When a valid reduction target is saved, the system displays the target and calculates progress using the recorded baseline/current emissions. Fail: the target is saved but progress is not updated after a new monthly calculation.	Supports the stated goal of tracking reduction milestones.
FR-005	Functional	The system shall allow a Sustainability Coach to review a Resident's recorded emissions history and target progress.	Medium	Pass: For an authorized Resident record, the Sustainability Coach can view monthly CO ₂ e history and current target progress. Fail: the coach cannot retrieve the required history or target status.	Enables the named Sustainability Coach stakeholder to support household sustainability decisions.
NFR-001	Non functional	The annual carbon reduction milestone chart shall render interactive historical comparison graphs in under 200 ms.	High	Pass: Benchmarking tests show the chart renders in less than 200 ms under the defined simulated peak-load test. Fail: the measured rendering time is 200 ms or more.	Provides responsive performance for the core tracking dashboard.
NFR-002	Non functional	The system shall protect household consumption and emissions records so that only the authorized Resident and assigned Sustainability Coach can access them.	High	Pass: An authorized user can retrieve the permitted record and an unauthorized user receives an access-denied response in 100% of security test cases. Fail: any unauthorized test case exposes household records.	Household utility and emissions data should not be exposed to unauthorized users.